Assessment 1: Mobile App Design Specification

Due Date: Friday Week 4 (22nd August, 11:55 PM) — Weight: 20%

Purpose:

The purpose of this assessment is to plan out the major iOS application that you will develop this semester.

Completion of this assessment should demonstrate the following unit learning outcomes:

- Describe the feasibility and technical challenges of creating iOS apps using UIKit and associated technologies
- Analyse mobile interface guidelines and technical constraints to design effective navigation and user interfaces for mobile apps
- Follow iOS best practices to design non-trivial iOS apps with a web service component.

Task:

For this assessment, you will produce a design specification for an innovative iOS app. Over the rest of the semester (as part of Assessment 3 and 4) you will develop the app you have described in this design document. This assessment is an opportunity to use creative and technical skills to plan an application of your own design or based on one of our suggestions and to get feedback on the feasibility of your design from your demonstrator.

> Before you begin working on your own design specification, you must get your lab demonstrator to approve your app idea, and the technologies you will use (below).

Feel free to ask on the Ed forums or approach any unit staff if you have any other queries regarding the assessment.

Assessment Options and Requirements:

For this assessment, you can either:

- Create an app completely of your own design, or
- Create an app based on one of our suggestions below

In either case, your design must utilise both the following two technologies:

- Persistent storage (Week 4 topics/file storage from Week 8),
- Web services (Week 5)

And must utilise at least two of the following technologies:

- Firebase Cloud Platform (Week 6)
- Maps and Location Services (Week 7)
- Audio / Video Frameworks (Week 8)
- Motion Services (Week 9)
- Swift Charts (Week 9)
- Gesture Handling (Week 9)
- Local Notifications (Week 10)
- Deep linking (Week 10)

You should consider these requirements before seeking approval from your demonstrator, however If you are stuck, they may be able to provide suggestions on how to meet them.

App Suggestions:

You may choose one of the following application ideas as a starting point. You must then expand this into your own unique take on the concept and have it approved by your demonstrator.

- Hobby Management App
  Create an application where people manage their desired hobby and keep track of their progress. This can include Blogging, Photos, Dates and Times, Planning, and ways to review their progress. The app should allow tracking of multiple types of hobbies.
- Cooking Assistant App
  Create an application to assist students with their daily cooking. This will include step-by-step guides, shopping list management, and recommendations. This app could connect to an API to source its content.
- Community of Interest App
  Create an application to allow people with a particular interest to find events of interest, organise meetings and communicate with local, like-minding people in their area. Make use of maps, chat, and an online data source (e.g., for locations). This could be centred around whatever interests you: dog parks, skateboarding, board gaming, etc.
- Tourist Guidance Application
  Create an application for helping people plan and manage their holidays, providing them with features such as notifications, recommendations, transportation, map integration, etc. The application should provide information on various elements within the application and be location-aware.

Marking Criteria:

Your design specification will be assessed on the following criteria (100 marks total). It can be useful to use these headings to structure your document.

Application Concept – 10 Marks

You must clearly articulate the application you wish to develop. Key things that must be included here are the target audience and a description of the app's features. You should have enough detail that the user can understand how your app will be used. It could be valuable to discuss the app concept with your peers or other potential users.

Competition and Innovation – 10 Marks

You should provide a competitor analysis, comparing your app to similar apps. You should review two published apps of the same type on the App Store and report on their key functionality, strengths, and weaknesses. You should clearly describe what makes your app innovative. You should discuss the key features that differentiate your application from existing apps of the same type. It is fine for your app to serve the same purpose as existing apps, but it must be better than existing offerings in some way.

Interface Design and Navigation – 40 Marks

Your specification needs to propose a draft user interface design and describe the navigation for the app. This should follow good usability practices. You must provide complete storyboards for the application and show the navigation hierarchy for the app. Detail is key here—the more information you can provide the better! The format in which you present your storyboards is up to you. They can be sketched or drawn with a professional tool, but they must be legible. Adobe XD or Lucid Charts can be useful.

Interface and Navigation Justification – 10 Marks

Your app's interface <u>must</u> be designed in line with Apple's Human Interface Guidelines (HIG). You should describe how elements of your app's interface design and navigation follow the advice in the HIG, giving references to the relevant sections.

Feasibility and Technology – 20 Marks

For your interface and navigation design, you should describe what iOS components, technologies, and frameworks can be used to build this. Imagine you are writing for a potential investor who doesn't know iOS and you are trying to convince them that your app idea is feasible. You should describe both the iOS frameworks/APIs/UI Elements that can or could be used. For aspects of the design you are unsure of (e.g., you can't determine if a certain iOS component provided the required functionality), you can describe these uncertainties.

You shouldn't depend only on what we have taught so far (see the unit outline in Week 1 for the things we cover). This <u>will require some additional research</u> from you to determine how you could implement specific functionality. You are not locked into these choices.

Scope and Estimated Project Timeline – 10 Marks

As you will be developing your app throughout the semester, you should ensure that your app's development scope and limitations are considered to ensure the project can realistically be completed. You should provide a breakdown of development tasks (different parts of the app) and describe the order they will be developed. Ideally this should be a list or timeline in terms of the prototype demonstrations (Assessment 3—Labs 8–12) and the final submission (Assessment 4). You should describe functionality necessary for the Minimum Viable Product (MVP), i.e., the minimum functionality necessary to have a useful app with the required technologies. You can detail desirable functionality that could be added if time permits.

Submission Requirements:

Your assessment will need to be submitted online via Moodle. Your design specification document must be submitted as a single PDF file. You should ensure that the file is named using the following convention STUDENTNAME-A1-MobileAppDesign.

Your submission should be logically structured and free of grammatical and spelling errors. Ensure that any external resources that you used to write your specification are referenced in the document. Failure to reference resources used can be considered plagiarism and result in zero marks being awarded for the assessment.

If you have any questions regarding any section of the assessment, feel free to post them on the unit Ed Forum.

Late Submission

Failure to submit your assessment on time will result in a 5% mark penalty for each day late (or part-day thereof) including weekends, up to a maximum of 7 days late. Submissions later than 7 days will not be marked.

Special Consideration

If you are unable to submit your assessment on time due to circumstances beyond your control, you may be eligible for special consideration. Special consideration applications must be submitted via Monash Connect, along with supporting documentation.

Generative AI tools may be used in guided ways within this assessment

You can use generative artificial intelligence (AI) to help with report writing (drafting and editing) and to conduct background research, however be aware it may not give correct answers or best practices, and may not give accurate sources. Given this assessment serves as a plan for what you will do throughout the rest of semester, it is beneficial for you to actively engage in the design process. Any use of AI must be clearly documented and acknowledged (see Learn HQ).